

# Sinan Özaydın

PhD, MSc, BS

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Sydney, New South Wales,  
Australia

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Github : [github.com/sinanozaydin](https://github.com/sinanozaydin)

## BIO

Earth scientist/Geophysicist specialising in magnetotellurics and geodynamic modeling. Sinan's research focuses on understanding tectonic and magmatic processes through interdisciplinary approaches that combine geophysical observations, numerical geodynamic modeling, petrophysical modelling, and petrological-geochemical data. He has developed two scientific software packages (pide, MATE) and contributed to the development of two others (SAnTex, Thermobar).

## EMPLOYMENT

**Postdoctoral Research Associate at University of Sydney**

*January 2023 -*

**Geophysics Consultant at Vox Geophysics**

*September 2024 -*

**Postdoctoral Research Fellow at Macquarie University** *January 2022 - December 2023*

**Postdoctoral Research Associate at Uni. of South Australia** *September - December 2021*

## EDUCATION

**PhD in Geophysics at Macquarie University**

*2018 - 2021*

Sydney, Australia

**Thesis Title:** *"Three-dimensional magnetotelluric constraints on compositional variations of the Southern African mantle."*

**Supervisors:** Kate Selway (Primary), William L. Griffin (Associate)

**MSc in Geophysics at Boğaziçi University**

*2015 - 2017*

Istanbul, Turkey

**Thesis Title:** *"The role of crustal fluids in tectonics of north-central Turkey, inferred from three-dimensional magnetotellurics."*

**Supervisors:** S. Bülent Tank

**BS. in Geophysical Engineering at Istanbul Technical University**

*2009-2015*

Istanbul, Turkey

## RESEARCH INTERESTS

Geophysics, magnetotellurics, geophysical inversion, petrology, geochemistry, numerical modelling, tectonics, geodynamic modelling, petrophysical interpretation.

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## PUBLICATIONS

18 - Han, Kui, **Özaydın, Sinan**, Fei, Hongzhan, Man, Lianjie, Wang, Fei, Chanyshv, Artem, Withers, Anthony C., Grayver, Alexander, Katsura, Tomoo (2026). “Lower mantle iron heterogeneity indicated by electrical conductivity”, *Science Advances*, 12(17), [10.1126/sciadv.aec7875](https://doi.org/10.1126/sciadv.aec7875).

17 - **Özaydın, Sinan**, Rey, Patrice F., Chatzaras, Vasileios (2026). “Strain controls the electrical conductivity distribution in the lithosphere”, *Geology*, 54 (2), [10.1130/G53957.1](https://doi.org/10.1130/G53957.1).

16 - Singh, Utpal, **Özaydın, Sinan**, Chatzaras, Vasileios, Rey, Patrice F. (2025) “SAnTex: A Python-based Library for Seismic Anisotropy Calculation”, *Journal of Open Source Software*, 10(110), 6886, [10.21105/joss.06886](https://doi.org/10.21105/joss.06886).

15 - **Özaydın, Sinan**, Li, L., Singh, U., Rey, P.F., Manassero, M.C (2025). “pide: Petrophysical Interpretation tools for geoDynamic Exploration.”, *Journal of Open Source Software*, 10(105), 7021, [10.21105/joss.07021](https://doi.org/10.21105/joss.07021).

14 - Manassero M.C., **Özaydın, Sinan**, Afonso, J.C., Shea, J., Kirkby, A., Ezad, I.S., Thiel, S., Fomin, I., Czarnota, K. (2024). “Lithospheric structure and melting processes in southeast Australia: new constraints from joint probabilistic inversions of 3D magnetotelluric and seismic data”, *Journal of Geophysical Research: Solid Earth*, 129, [10.1029/2023JB028257](https://doi.org/10.1029/2023JB028257).

13 - **Özaydın, Sinan**, Selway, Kate, Foley, Stephen F., Ezad, Isra S., Griffin William L. Tarits, Pascal, Hautot, Sophie (2024). “Role of metasomatism in the development of the East African Rift at the Northern Tanzanian Divergence: Insights from 3D magnetotelluric modelling.”, *Geochemistry, Geophysics, Geosystems*, 25, [10.1029/2023GC011191](https://doi.org/10.1029/2023GC011191).

12 - Selway, Kate, **Özaydın, Sinan**, Payne, Justin (2023). “Metasomatism and depletion of the southern Gawler Craton from combined mantle xenocryst and AusLAMP magnetotelluric data”, *Exploration Geophysics*, 1-15, [10.1080/08123985.2023.2282711](https://doi.org/10.1080/08123985.2023.2282711)

11 - Han, Kui, Guo, Xinzhuang, Wang, Xuben, Zhang, Junfeng, **Özaydın, Sinan**, Li, Dewei, Clark, Simon Martin (2023). “The electrical conductivity of granite: The role of hydrous accessory minerals and the structure water in major minerals.”, *Tectonophysics*, 229857, 856. [10.1016/j.tecto.2023.229857](https://doi.org/10.1016/j.tecto.2023.229857).

10 - Wieser P., Petrelli M., Lubbers J., Wieser E., **Özaydın, Sinan**, Kent A., Till C., (2022). “Thermobar: Open source thermobarometry and hygrometry in Python3.”, *Volcanica*, 5(2), [doi:10.30909/vol.05.02.349384](https://doi.org/10.30909/vol.05.02.349384).

9 - **Özaydın, Sinan**, Selway, Kate (2022). “The Relationship Between Kimberlitic Magmatism and Electrical Conductivity Anomalies in the Mantle.”, *Geophysical Research Letters*, 49, e2022GL099661, [doi:10.1029/2022GL099661](https://doi.org/10.1029/2022GL099661)

8 - **Özaydın, Sinan**, Selway, Kate, Griffin, William L., Moorkamp, Max (2022). “ Probing the southern African lithosphere with magnetotellurics: 2. Linking electrical conductivity, composition, and tectonomagmatic evolution.”, *Journal of Geophysical Research: Solid Earth*, 127, e2021JB023105. [doi:10.1029/2021JB023105](https://doi.org/10.1029/2021JB023105)

7 - Moorkamp, Max, **Özaydın, Sinan**, Selway, Kate, Jones, Alan G., (2022). “Probing the Southern African Lithosphere With Magnetotellurics—Part I: Model Construction”, *Journal of Geophysical Research: Solid Earth*, 127, e2021JB023117. [doi:10.1029/2021JB023117](https://doi.org/10.1029/2021JB023117)

6 - **Özaydın, Sinan**, Selway, Kate, Griffin, William L. (2021). “Are xenoliths from southwestern Kaapvaal Craton representative of the broader mantle? Constraints from magnetotelluric modeling”, *Geophysical Research Letters*, 48, [doi:10.1029/2021GL092570](https://doi.org/10.1029/2021GL092570).

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5 - **Özaydın, Sinan**, Selway, Kate, “MATE: An analysis tool for the interpretation of magnetotelluric models of the mantle” (2020). *Geochemistry, Geophysics, Geosystems*, 21, doi:10.1029/2020gc009126.

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4 - Selway, Kate, O'Donnell, J. P., **Özaydın, Sinan** (2019). “Upper Mantle Melt Distribution From Petrologically Constrained Magnetotellurics”, *Geochemistry, Geophysics, Geosystems*, 20, doi:10.1029/2019GC008227.

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3 - Tank, S. Bülent, **Özaydın, Sinan**, Karas, Mustafa (2018). “Revealing the electrical properties of a gneiss dome using three-dimensional magnetotellurics: Burial and exhumation cycles associated with faulting in Central Anatolia, Turkey”, *Physics of the Earth and Planetary Interiors*, 283, doi:10.1016/j.pepi.2018.07.010.

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2 - **Özaydın, Sinan**, Tank, S. Bülent, Karas, Mustafa (2018). “Electrical resistivity structure at the North-Central Turkey inferred from three-dimensional magnetotellurics.”, *Earth, Planets and Space*, 70(49), doi:10.1186/s40623-018-0818-4.

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1 - Karas, Mustafa, Tank, S. Bülent, **Özaydın, Sinan** (2017). “Electrical conductivity of a locked fault: investigation of the Ganos segment of the North Anatolian Fault using three-dimensional magnetotellurics.”, *Earth, Planets and Space*, 69(107), doi:10.1186/s40623-017-0695-2.

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PUBLICATIONS  
IN PROGRESS

Li, P., Chatzaras, V., Foley, M., **Özaydın, Sinan**, Rey, P. (2026). “A standardized workflow for quantitative 3D shape preferred orientation analysis of minerals using X-ray computed tomography”, PLOS One, Under Review.

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Rychert, Catherine A. , Dai, Yuhang **Özaydın, Sinan**, Debayle, Eric, Harmon, Nicholas, Chin, Emily J., Conrad, Clinton P., Hirth, Greg, Naif, Samer, Selway, Kate, Artemieva Irina (2026). “Variations and dynamics of the lithosphere-asthenosphere boundary”, *Seismica*, Under Review.

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PUBLICATIONS  
IN PREPARATION

Al-Qahtani, Yahya, **Özaydın, Sinan**, Chatzaras, Vasileios, Rey, Patrice F. (2025). “Electrical conductivity structure of Broken Hill and its relationship to Pb-Zn mineralization”, *Geochemistry, Geophysics, Geosystems*, in preparation.

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Singh, Utpal, Chatzaras, Vasileios, **Özaydın, Sinan**, Rey, Patrice F. (2025). “Anisotropic structure beneath North Anatolian Fault.”, *Geophysical Research Letters*, in preparation.

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**Özaydın, Sinan**, Rey, Patrice F., Chatzaras, Vasileios, Hoggard, Mark J. (2025). “Why intraplate earthquakes avoid mantle conductors? A new way to constrain seismic hazard in intra-continental settings.”, *Earth and Planetary Science Letters*, in preparation.

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**Özaydın, Sinan**, Li, Lu, Rey, Patrice F., Chatzaras, Vasileios, Selway, Kate (2026). “A stochastic mantle composition characterization scheme using magnetotellurics and seismic tomography: An application in Southeast Australia.”, *Geophysical Journal International*, in preparation.

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**Özaydın, Sinan**, Munch, Federico, Grayver, Alexander (2026). “Compositional heterogeneity of lithospheric mantle beneath US through petrological-geophysical interpretation”, *To be determined*, in preparation.

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|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REVIEWED<br>MANUSCRIPTS IN                                    | Nature; Geology; Nature Communications; Nature Communications Earth & Environment; Science Advances; Geophysical Journal International; Geophysical Research Letters, Journal of Open Source Software; Gondwana Research; Journal of Geophysical Research: Solid Earth; Geochemistry, Geophysics, Geosystems; Physics of the Earth and Planetary Interiors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| RESEARCH<br>PROJECTS                                          | <p><b>ARC-DP Identifying mineral systems by geochemical, geophysical and geological mapping of deep Australia</b> 2024 -<br/>Research Associate</p> <hr/> <p><b>ARC-Linkage Project Evolution of Proterozoic multistage rift basins – key controls on mineral systems</b> 2023 - 2024<br/>Research Associate</p> <hr/> <p><b>Compositional constraints on the evolution of Tanzanian Craton and surrounding rift margins using magnetotelluric modelling.</b> 2022 - 2023<br/>Researcher</p> <hr/> <p><b>Unveiling the relationship between kimberlites and electrical conductivity signatures of the mantle.</b> 2021 - 2022<br/>Researcher</p> <hr/> <p><b>Improving the quantitative interpretations of magnetotelluric models of the cratonic mantle.</b> 2018 - 2022<br/>Researcher / PhD Candidate</p> <hr/> <p><b>Summit Station Magnetotellurics (SUMMAT)</b> 2018<br/>Researcher</p> <hr/> <p><b>Continental Dynamics / Central Anatolian Tectonics (CD-CAT)</b> 2014-2018<br/>Intern / Researcher</p> <hr/> <p><b>Imaging the Shallow Crustal Structure of Ganos Fault by Magnetotellurics</b> 2015-2017<br/>Researcher</p>                                                                         |
| CONFERENCE<br>PROCEEDINGS<br>(1 <sup>st</sup> AUTHOR<br>ONLY) | <p><b>Özaydın, Sinan</b>, Patrice F. Rey, Vasileios Chatzaras (2024). “Thermomechanical modelling as a tool for decoding geophysical anomalies: A case study of a strike-slip pull-apart basin.”, <i>SGTSG2024</i>, Armidale, Australia.</p> <hr/> <p><b>Özaydın, Sinan</b>, Patrice F. Rey (2024). “Entering the anomaly factory: Conversion of thermomechanical geodynamic models into magnetotelluric models. An example from a pull-apart basin under a strike-slip stress regime.”, <i>EMIW2024</i>, Beppu, Japan.</p> <hr/> <p><b>Özaydın, Sinan</b>, Lu Li, Utpal Singh, Patrice F. Rey, M.C. Manassero (2024) “pide: a python-based petrophysical interpretation tool of electrical conductivities of materials from crust to mantle transition zone.”, <i>EMIW2024</i>, Beppu, Japan.</p> <hr/> <p><b>Özaydın, Sinan</b> (2023). “Understanding the compositional nature and architecture of the lithosphere using magnetotellurics. University of Sydney, Earthbyte Seminar Series.</p> <hr/> <p><b>Özaydın, Sinan</b>, Rey, Patrice F., Selway, Kate, Giordani, Julian (2023). “Magnetotelluric insights in to the rheology and composition of the mantle and applications for geodynamic mod-</p> |

elling.” Oral Presentation, *AESC2023*, Perth, Australia.

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**Özaydın, Sinan**, Selway, Kate, Foley, Stephen F., Tarits, Pascal, Sophie, Griffin, William L., Ezad, Isra S. (2023). “Unveiling the compositional nature and architecture of the lithospheric mantle at Northern Tanzanian Divergence with magnetotellurics.”, Poster Presentation, *AESC2023*, Perth, Australia.

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**Özaydın, Sinan**, Selway, Kate, Moorkamp Max, Griffin, William L., Manassero, M. C. (2022). “What are the compositional causes behind electrical conductivity variations in continental lithospheric mantle? Methodology and practice for quantified interpretations.”, Poster Presentation, *EMIW2022*, Çeşme, Turkey.

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**Özaydın, Sinan**, Selway, Kate, Foley, Stephen F., Tarits, P., Hautot, S. (2022). “Investigation into lithospheric mantle of Northern Tanzania utilising 3D magnetotellurics.”, Poster Presentation, *EMIW2022*, Çeşme, Turkey.

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**Özaydın, Sinan**, Selway, Kate “Laboratory results coded in MATE.”, Oral Presentation (Invited Speaker), *EM-Community Webinar Series*, Virtual Conference, International.

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**Özaydın, Sinan**, Selway, Kate, Griffin, William L. (2021). “Composition and evolution of the southern African lithosphere from combined xenocryst and magnetotelluric data”, Oral Presentation, *AESC2021*, Virtual Conference, Australia.

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**Özaydın, Sinan**, Selway, Kate (2020). “Interpretation of conductivity variations in magnetotelluric models of cratonic lithospheric mantle with the new open-source software MATE.”, Oral Presentation, *AGU2020*, Virtual Conference, USA.

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**Özaydın, Sinan**, Selway, Kate (2019). “Utilising 3-D magnetotelluric models of southern African mantle to constrain hydrogen content and compositional variations.”, Oral Presentation, *AEGC2019*, Perth, Australia.

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**Özaydın, Sinan**, Selway, Kate (2018). “Measuring the hydrogen content variations in Southern African mantle.”, Poster Presentation, *EMIW2018*, Helsingör, Denmark.

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**Özaydın, Sinan**, Tank, S. Bülent, Karas, Mustafa, Sandvol, Eric (2017). “Resolving the deep electrical resistivity structure at Central Pontides, Northern Turkey by three-dimensional magnetotelluric modeling.”, Poster Presentation, *EGU2017*, Vienna, Austria.

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## GRANTS

University of Sydney CAPEX Funding Scheme for MT instruments - 200.000AUD (2023)  
NCI Adapter Scheme Q3 - 250 KSU Computing Time at Gadi Supercomputer (2023)  
Macquarie University Covid Recovery Postdoctoral Fellowship - Salary + 5000AUD (2021-2022)

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## AWARDS & ACHIEVEMENTS

Australian Research Council PhD research scholarship 2018-2021  
TUBITAK research scholarship 2016-2017

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SCIENTIFIC  
SOFTWARE

**pide** - Creator - <https://github.com/sinanozaydin/pide>

A python library to make petrophysical calculations of earth materials from crust to mantle transition zone.

**MATE** - Creator - <https://github.com/sinanozaydin/MATE>

A software to make quantified interpretations of the magnetotelluric models of the mantle. Published in G-cubed.

**SAnTex** - Contributor - <https://github.com/utpal-singh/SAnTex>

A python library to calculate elastic properties of rocks.

**Thermobar** - Contributor - <https://github.com/PennyWieser/Thermobar>

A python library to handle thermobarometric calculations of earth materials.

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FIELDWORK  
EXPERIENCE

**Broken Hill**

*2024*

Installation of BBMT stations around Broken Hill.

**Greenland Summit Station Fieldwork**

*2018*

Installation of broad-band and long-period MT stations in Greenland ice sheet.

**CD-CAT Fieldwork in Central Anatolia**

*2014-2018*

Installation of nearly 150 broad-band and long-period MT stations in Central Anatolia.

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HIGHER DEGREE  
RESEARCH  
SUPERVISION

**Utpal Singh** - PhD co-advisor (2023 -), University of Sydney.

**Yahya Al-Qahtani** - PhD co-advisor (2023 - ), University of Sydney.

**Puqing Li** - PhD co-advisor (2023 -), University of Sydney.